MATH 243 — Probability & Statistics

Previous-year question analysis and study priority guide

What was analysed

12 BUET L-2/T-1 & L-2/T-2 CSE papers (2011-12 to 2023-24): the statistics/probability parts of the older MATH 241 / MATH 245 "Complex Variables and Statistics" papers (2011-12 to 2021-22), plus the two papers of the current course, **MATH 243 (2022-23 and 2023-24, your 22-batch paper)**, where all 8 questions are probability & statistics. Complex-variable questions were excluded. The 2018-19 paper appears twice in your merged file and was counted once. Every sub-question was classified into one of 16 topics that match your course slides (12 stats lectures + 12 probability lectures) and its marks were summed.

Key findings at a glance

• **Hypothesis testing** (z-test, t-test, proportions, confidence intervals, Type I/II errors) is the single biggest topic: asked in **all 12 papers**, 253 marks in total, and 82 marks across the two MATH 243 papers (≈ 1 full question + per paper).

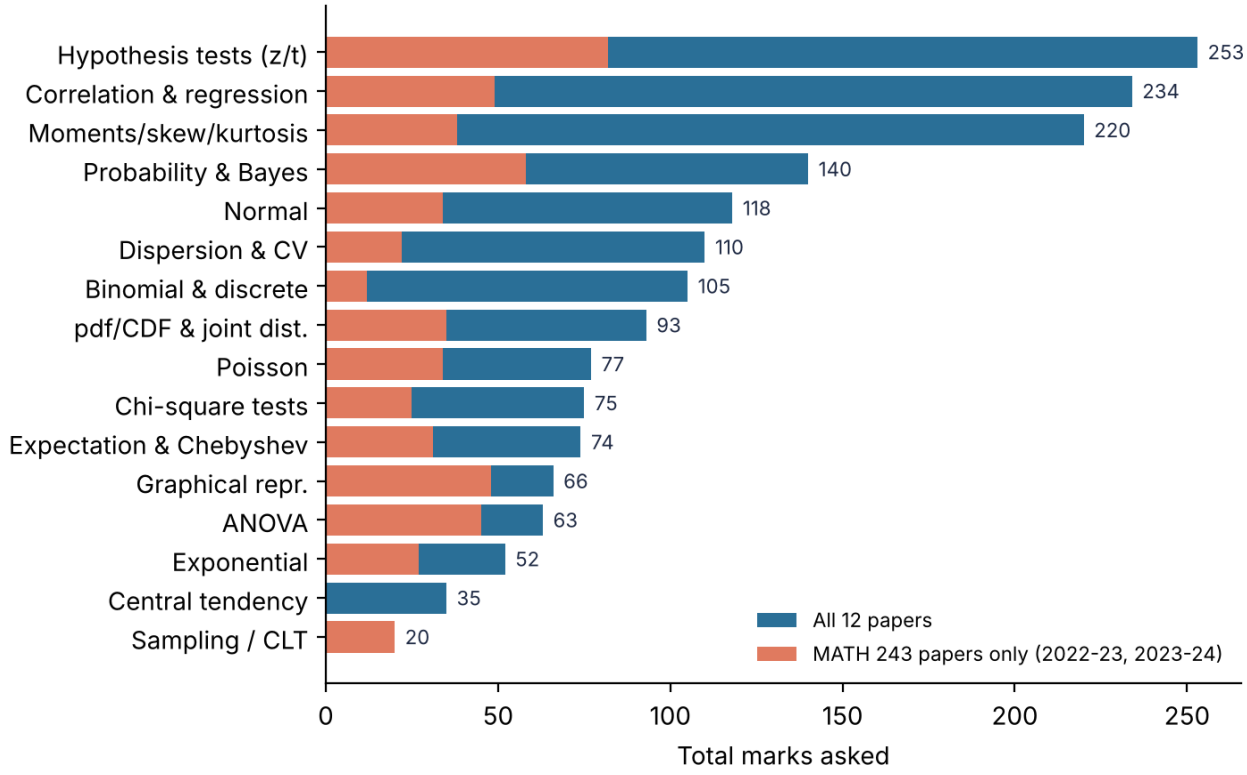
• **Correlation & regression** (12/12 papers, 234 marks) and **moments, skewness & kurtosis** (12/12 papers, 220 marks) are guaranteed every year. They are also the most "mechanical" topics — big marks for a fixed procedure.

• In the new MATH 243 format **Section A = probability** (Q1 probability/Bayes, Q2 random variables & joint distributions, Q3 discrete distributions + Chebyshev, Q4 normal + correlation/regression) and **Section B = statistics** (Q5 descriptive/graphs, Q6 moments + sampling/CLT, Q7 hypothesis tests, Q8 chi-square + ANOVA). The two MATH 243 papers follow this layout almost identically — expect it again.

• **New topics since MATH 243 began:** ANOVA (asked in 3 of the last 3 papers, 20–25 marks each), chi-square contingency tables, stem-and-leaf / scatter / histogram questions, CLT & sampling, rank correlation and Chebyshev's theorem. Older papers barely had these, so do not under-weight them just because their all-time total is lower.

• **Gap in the slides:** none of your stats slides is a dedicated lecture on z/t tests, chi-square tests or confidence intervals (only the ANOVA lecture covers the hypothesis-testing framework). Yet this is the highest-mark area. Study it from the textbook (Walpole) or class notes as well.

1. Which topic gets the most questions?

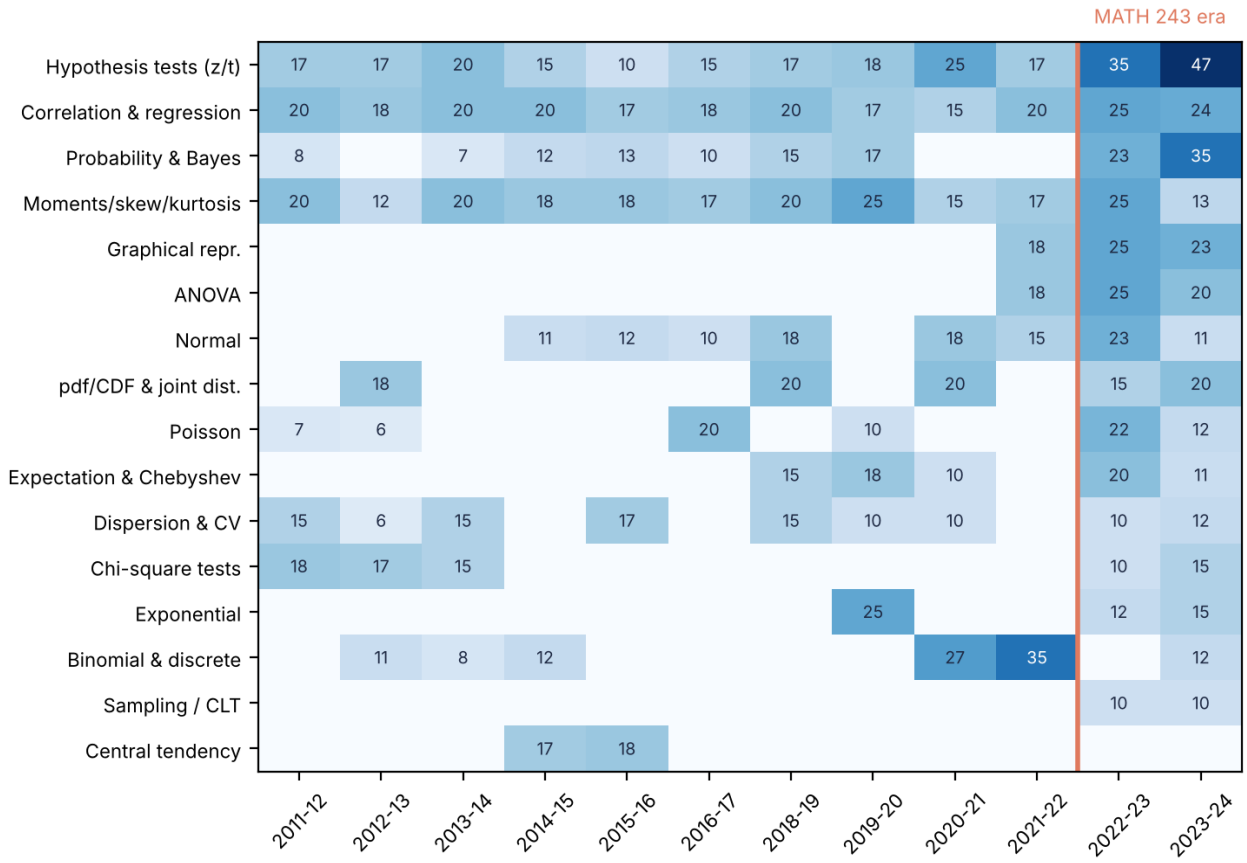


Orange part of each bar = marks in the two MATH 243 papers only. Topics with large orange share (ANOVA, graphical representation, hypothesis tests, probability) are the ones the current examiners favour.

#	Topic	Papers (of 12)	Sub-questions	Total marks	MATH 243 marks (2 papers)	% of all marks
1	Hypothesis testing: z / t tests, proportions, CI, concepts	12	12	253	82	14.6%
2	Correlation & regression (incl. rank correlation, R^2)	12	12	234	49	13.5%
3	Moments, skewness & kurtosis	12	13	220	38	12.7%
4	Basic probability, conditional, total probability & Bayes	9	9	140	58	8.1%
5	Normal distribution	8	8	118	34	6.8%
6	Dispersion & coefficient of variation	9	9	110	22	6.3%
7	Binomial & other discrete (hypergeometric, multinomial, geometric)	6	8	105	12	6.1%
8	Random variables: pdf/CDF & joint distributions	5	6	93	35	5.4%
9	Poisson distribution	6	6	77	34	4.4%
10	Chi-square tests (goodness of fit, contingency)	5	5	75	25	4.3%
11	Expectation, variance, covariance & Chebyshev	5	5	74	31	4.3%
12	Graphical representation (stem-leaf, histogram, ogive, scatter)	3	4	66	48	3.8%
13	ANOVA (one-way & two-way)	3	3	63	45	3.6%

#	Topic	Papers (of 12)	Sub-questions	Total marks	MATH 243 marks (2 papers)	% of all marks
14	Exponential distribution	3	3	52	27	3.0%
15	Central tendency (mean/median/mode, missing frequencies)	2	2	35	0	2.0%
16	Intro / sampling / CLT	2	2	20	20	1.2%

2. Year-by-year pattern



Marks per topic per paper. Older papers (left of the orange line) had only 3–4 stats questions; the MATH 243 papers (right) have 8. Rows are sorted by the priority score used in section 3.

Patterns worth noticing

• Regression line(s) + correlation coefficient appeared every single year, usually for 15–25 marks, often with "estimate Y when X = ...".

• Skewness/kurtosis from a grouped frequency table (moments about assumed mean → central moments → β_1, β_2 / γ_1, γ_2) appeared every year. The *same* table (10-20 ... 50-60 with frequencies 18, 20, 30, 22, 10) was reused in 2014-15, 2016-17, 2018-19 and 2022-23.

• Questions repeat almost word-for-word: GPA z-test with 99% CI (2018-19 and 2022-23); "640/320 families with 5 children" chi-square (2012-13, 2013-14); mice life-span (2019-20, 2020-21); missing frequencies with median 413.11/415 (2014-15, 2015-16). Solving every past paper is very high-value.

• Comparing two groups with the coefficient of variation ("which is more consistent / more variable / more stable") is asked in 9 of 12 papers.

• Section A of MATH 243 always opens with a two-box/two-bag transfer problem (total probability) followed by a Bayes' theorem problem.

3. What to study first — priority ranking

Priority score = share of all-time marks + 2 × share of MATH 243 marks (recent papers count double, because they reflect the current syllabus and examiners). Tiers:

Rank	Topic	Tier	Score	Slides to use
1	Hypothesis testing: z / t tests, proportions, CI, concepts	A — must master	43.9	ANOVA lecture (hypothesis framework) + textbook; t-table / normal table files
2	Correlation & regression (incl. rank correlation, R^2)	A — must master	31.0	Stats 7: Correlation & regression
3	Basic probability, conditional, total probability & Bayes	A — must master	28.8	Prob L2, L3&4, L5
4	Moments, skewness & kurtosis	A — must master	26.3	Stats 4: Skewness, Moments & Kurtosis
5	Graphical representation (stem-leaf, histogram, ogive, scatter)	A — must master	20.9	Stats 5 & 6: Graphical representation
6	ANOVA (one-way & two-way)	B — high value	19.7	Stats 8: ANOVA + F-table
7	Normal distribution	B — high value	18.9	Prob L12 + standard normal table
8	Random variables: pdf/CDF & joint distributions	B — high value	17.9	Prob L6, L7, L8
9	Poisson distribution	B — high value	16.6	Prob L11
10	Expectation, variance, covariance & Chebyshev	B — high value	15.3	Prob L9 + "other distributions" (Chebyshev)
11	Dispersion & coefficient of variation	B — high value	14.2	Stats 3: Measures of Dispersion
12	Chi-square tests (goodness of fit, contingency)	C — secure the easy marks	13.3	StatisticalTables (χ^2) + textbook
13	Exponential distribution	C — secure the easy marks	12.6	"Other distributions" lecture
14	Binomial & other discrete (hypergeometric, multinomial, geometric)	C — secure the easy marks	10.3	Prob L10 + "other distributions"
15	Intro / sampling / CLT	C — secure the easy marks	8.3	Stats 1.1, 1.2 + textbook (CLT)
16	Central tendency (mean/median/mode, missing frequencies)	C — secure the easy marks	2.0	Stats 2: Central Tendency

Recommended study order

Order	Topic	What exactly to practise
Step 1	Hypothesis testing	One-sample z and t tests, two-sample t, two-proportion z test, confidence intervals, P-value, Type I/II errors and the definitions (critical region, one/two-tailed, level of significance). ≈ 35–47 marks in each MATH 243 paper.
Step 2	Correlation & regression	Both regression lines, r, regression coefficients, prediction, rank correlation (with ties), SSR/SST/R^2, second-degree polynomial fit.
Step 3	Probability, total probability & Bayes	Bag-transfer problems, Bayes with 3 machines/parties, independence vs mutually exclusive, reliability circuits. Always Q1 of Section A.
Step 4	Descriptive statistics: moments/skewness/kurtosis, CV, graphs	One long grouped-table moments calculation (practise until fast and error-free), CV comparison, corrected mean/SD, plus stem-and-leaf, scatter plot, histogram/ogive with Sturges' rule (23 marks in 2023-24).

Order	Topic	What exactly to practise
Step 5	Normal distribution	z-area problems, finding μ/σ from percentages, "how many units…", mode/properties proof.
Step 6	ANOVA + chi-square	One-way ANOVA (every recent paper), two-way ANOVA, contingency-table χ^2 and goodness-of-fit. Usually the whole of Q8.
Step 7	Random variables, joint distributions & expectation	Find k, CDF ↔ pdf, marginal/conditional, discrete joint table with covariance, E[g(X)], Var of aX+b, Chebyshev.
Step 8	Binomial, Poisson, exponential & others	Binomial–Poisson relation proof, Poisson approximation, memoryless exponential, hypergeometric, multinomial.
Step 9	Central tendency, sampling/CLT	Missing frequencies with median/mode, CLT theory and its CS applications, stratified sampling, sources of statistical data — short, scoring theory parts.

4. Topic-by-topic: how each is asked

Hypothesis testing: z / t tests, proportions, CI, concepts

Seen in exams: "Can this sample be regarded as from a population with mean 47?" (t, 2011-12); talcum tins t-test (2013-14); bulb longevity z (2014-15); chocolate-bar advertising t (2015-16); life-span z (2016-17); GPA z + 99% CI (2018-19, 2022-23); mice P-value (2019-20); sodium t + Type I/II paternity (2020-21); heights t (2021-22); bank waiting time t (2022-23); delivery time z, two-proportion test, two-sample t (2023-24).

Tip: Know which test to pick: σ known or n ≥ 30 → z; small n, σ unknown → t (df = n−1 or n_1+n_2-2). Always write H_0, H_1, test statistic, critical value, decision, conclusion.

Correlation & regression (incl. rank correlation, R^2)

Seen in exams: Two regression lines + r from raw X, Y data almost every year; from summary stats (means, SDs, r) in 2018-19; SSR/SST/R^2 in 2016-17 and 2020-21; rank correlation in 2022-23 and 2023-24; quadratic fit + R^2 in 2023-24.

Tip: Use the tabular method (Σx, Σy, Σxy, Σx^2, Σy^2) with coded values to save time; remember $b_{yx} \cdot b_{xy} = r^2$.

Moments, skewness & kurtosis

Seen in exams: Pearson's / Bowley's / Kelly's coefficients, moments about assumed mean → central moments, β_1, β_2, γ_1, γ_2, and "comment on the nature of the distribution".

Tip: Memorise μ_2, μ_3, μ_4 conversion formulas and interpretation ($\beta_2 = 3$ mesokurtic etc.).

Basic probability, conditional, total probability & Bayes

Seen in exams: Bag transfer (2022-23, 2023-24), Bayes with machines/parties/truth serum, cards, picnic sunburn/mosquito, couple watching TV, circuit reliability, independent solvers.

Tip: Draw a tree diagram; check whether events are independent.

Normal distribution

Seen in exams: Units under a weight, tube-light replacement interval, marks with 10% below 65 / 25% above 75, ball-bearing tolerance, trip-time percentiles, U-shaped component (sum of normals).

Tip: Practise reverse look-ups (given area → z).

ANOVA (one-way & two-way)

Seen in exams: One-way: salesmen sales (2021-22), electric bulb brands (2022-23), city prices (2023-24). Two-way: corn types × blocks (2022-23).

Tip: Compute SST, SSB(SSTr), SSE, df, MS, F and compare with the given F value.

Chi-square tests (goodness of fit, contingency)

Seen in exams: Coins/families goodness-of-fit (binomial expected frequencies), drug vs typhoid 2×2, education vs job performance 3×3.

Tip: Expected = row total × column total / grand total; df = (r−1)(c−1).

Random variables, joint distributions & expectation

Seen in exams: Find k for valid pdf, P(a<X<b), CDF → pdf, joint density marginal & conditional, discrete joint table (pens), E[Y] for Y = 3X−2 or X+4, Chebyshev comparison.

Tip: Always check $\int f = 1$ first; $\text{Var}(aX+b) = a^2\text{Var}(X)$.

Binomial, Poisson & exponential

Seen in exams: Binomial probabilities (truck tyres, recovery, graduates), Poisson approximation (insurance claims, diode failures, tax errors, bubbles), derive Poisson / binomial→Poisson limit, exponential calls with memoryless property.

Tip: Be ready for the binomial→Poisson derivation — asked 3 times as a theory part.

5. Exam strategy

• You answer 3 of 4 questions per section. If short on time, you can safely leave the weakest of: discrete distributions / graphs — but never hypothesis testing, regression or probability/Bayes.

• Bring speed to long table calculations (moments, regression, ANOVA): these give 15–25 marks each and are pure procedure.

• Do not skip short theory parts (definitions of statistic vs parameter, Type I/II, CLT applications in CS, stem-and-leaf, sources of statistical data) — easy 6–10 marks.

• Solve the 2022-23 and 2023-24 MATH 243 papers fully under timed conditions; they are the best model of your exam.